

### Exercise 3 Mhlun Maswanganyi

① The products table

```
SELECT product_name,  
       price  
CASE  
    WHEN price > 1000 THEN 'Expensive'  
    WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'  
    WHEN price < 100 THEN 'Budget'  
END AS price_category  
FROM products;
```

② ~~SELECT customer\_name,~~

| <del>amount,</del>     | product_name | price   | price_category |
|------------------------|--------------|---------|----------------|
| <del>CASE</del>        | Laptop       | 1200.00 | Expensive      |
| <del>WHEN amount</del> | Phone        | 800.00  | Mid-range      |
|                        | Keyboard     | 45.00   | Budget         |
|                        | Monitor      | 300.00  | Mid-range      |
|                        | Mouse        | 25.00   | Budget         |

③ The orders table

```
SELECT customer_name,  
       amount,  
CASE  
    WHEN amount ≥ 1000 THEN 'High Value'  
    WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'  
    WHEN amount < 500 THEN 'Low Value'  
END AS order_value_category  
FROM orders;
```

③ The employees table

| customer-name | amount  | order_value_category |
|---------------|---------|----------------------|
| Alice         | 150.00  | Low value            |
| Bob           | 560.00  | Medium value         |
| Charlie       | 999.99  | Medium value         |
| Diana         | 45.50   | Low value            |
| Ethan         | 1200.00 | High value           |

SELECT emp-name,  
depart

③ The employees Table

SELECT emp\_name,  
department,  
salary

CASE

WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'

WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'

ELSE 'Staff'

FROM employees; END AS position\_level

FROM employees;

| emp_name | department | Salary | position_level       |
|----------|------------|--------|----------------------|
| John     | IT         | 85000  | Senior IT            |
| Sara     | HR         | 60000  | Staff Experienced HR |
| Mark     | IT         | 75000  | Staff                |
| Lucy     | Finance    | 95000  | Staff                |
| Tom      | HR         | 55000  | Staff                |

④ SELECT student\_name,  
Score,

CASE

WHEN score ≥ 90 THEN 'A'

WHEN score BETWEEN 80 AND 89 THEN 'B'

WHEN score BETWEEN 70 AND 79 THEN 'C'

WHEN score BETWEEN 60 AND 69 THEN 'D'

WHEN score < 60 THEN 'F'

END AS grade  
FROM students;

| Student | Score | grade |
|---------|-------|-------|
| Anna    | 92    | A     |
| Ben     | 76    | C     |
| Caru    | 59    | F     |
| David   | 83    | B     |
| Ella    | 68    | D     |

```
⑤ SELECT delivery-id,  
        delivery-time-minutes,  
        CASE  
            WHEN delivery-time-minutes ≤ 30 THEN 'Fast'  
            WHEN delivery-time-minutes BETWEEN 31 AND 60 THEN 'On Time'  
            WHEN delivery-time-minutes > 60 THEN 'Late'  
        END AS performance  
FROM deliveries;
```

| delivery-id | delivery-time-minutes | Performance |
|-------------|-----------------------|-------------|
| 1           | 45                    | on Time     |
| 2           | 80                    | Late        |
| 3           | 30                    | Fast        |
| 4           | 65                    | Late        |
| 5           | 100                   | Late        |



⑥ SELECT issue-type,  
priority,  
CASE

WHEN priority = '3' THEN 'High'

WHEN priority = '2' THEN 'Medium'

WHEN priority = '1' THEN 'Low'

END AS priority-level

FROM tickets

| ticket-id | issue-type     | Priority | priority-level |
|-----------|----------------|----------|----------------|
| 1         | Login issue    | 1        | Low            |
| 2         | Server down    | 3        | High           |
| 3         | Slow system    | 2        | Medium         |
| 4         | Email error    | 2        | Medium         |
| 5         | Password reset | 1        | Low            |

⑦ SELECT student-id,

~~days-present~~, (days-present \* 100 / total-days) AS  
~~total-days~~, attendance-percentage,

CASE

WHEN (days-present \* 100 / total-days)  $\geq$  90 THEN 'Excellent'

WHEN (days-present \* 100 / total-days) BETWEEN 75 AND 89.99  
THEN 'Good'

ELSE 'Needs Improvement'

END AS attendance-status

FROM attendance;

| student_id | attendance_percentage | attendance_status |
|------------|-----------------------|-------------------|
| 1          | 90.00                 | Excellent         |
| 2          | 60.00                 | Needs Improvement |
| 3          | 96.00                 | Excellent         |
| 4          | 50.00                 | Needs Improvement |
| 5          | 100.00                | Excellent         |

```

(88) SELECT product_id,
        stock_qty,
        CASE
            WHEN stock_qty = '0' THEN 'out of stock'
            WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low stock'
            WHEN stock_qty > 5 THEN 'In Stock'
        END AS stock_status
FROM products_inventory;

```

| Product_id | stock_qty | products_inventory |
|------------|-----------|--------------------|
| 1          | 5         | Low stock          |
| 2          | 0         | out of stock       |
| 3          | 25        | In stock           |
| 4          | 10        | In stock           |
| 5          | 3         | Low stock          |

```

(89) SELECT subject,
        enrolled_students,
        CASE
            WHEN enrolled_students ≥ 25 THEN 'Large'
            WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
            WHEN enrolled_students < 10 THEN 'Small'
        END AS class_size_category
FROM classes;

```

| Class-id | Subject | enrolled-student | class-size-category |
|----------|---------|------------------|---------------------|
| 1        | Math    | 30               | Large               |
| 2        | English | 25               | Large               |
| 3        | Science | 15               | Medium              |
| 4        | Art     | 5                | Small               |
| 5        | History | 20               | Medium              |

```

⑩ SELECT payment_id,
        payment_method,
        amount,
        CASE
            WHEN payment_method = 'cash' AND amount ≥ 200 THEN
                'Eligible for discount'
            ELSE 'Not Eligible'
        END AS discount_eligibility
FROM payments;

```

| payment_id | payment_method | amount | discount_eligibility  |
|------------|----------------|--------|-----------------------|
| 1          | Card           | 50.00  | Not eligible          |
| 2          | Cash           | 200.00 | Eligible for discount |
| 3          | Card           | 150.00 | Not eligible          |
| 4          | Paypal         | 25.00  | Not eligible          |
| 5          | Cash           | 200.00 | Eligible for discount |